

EXPLORING AI DISCUSSIONS ON BLUESKY

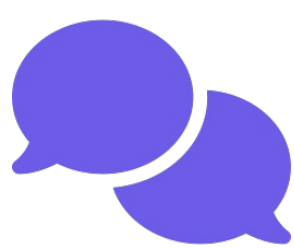
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Computer, Data and
Mathematical Sciences

Topic clustering, user engagement and social network analysis

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1 **RQ1**
WHAT ARE THE MAIN TOPICS IN AI-RELATED POSTS ON BLUESKY?

2 **RQ2**
DOES USER ENGAGEMENT DIFFER ACROSS AI DISCUSSION TOPICS?

3 **RQ3**
HOW ARE AUTHORS CONNECTED THROUGH FOLLOW RELATIONSHIPS, AND WHICH ACCOUNTS ARE CENTRAL?



DATA

Public Bluesky posts via
atrrr, 7–8 Oct 2026



SEARCH TERMS

artificial intelligence ·
machine learning ·
ChatGPT · generative AI



TEXT PROCESSING

cleaning · stemming · TF-IDF



ANALYSIS

k-means (RQ1) ·
randomisation test (RQ2) ·
follow network (RQ3)

1,168

posts collected

1,031

usable text posts

887

unique authors

5,483

TF-IDF terms

7

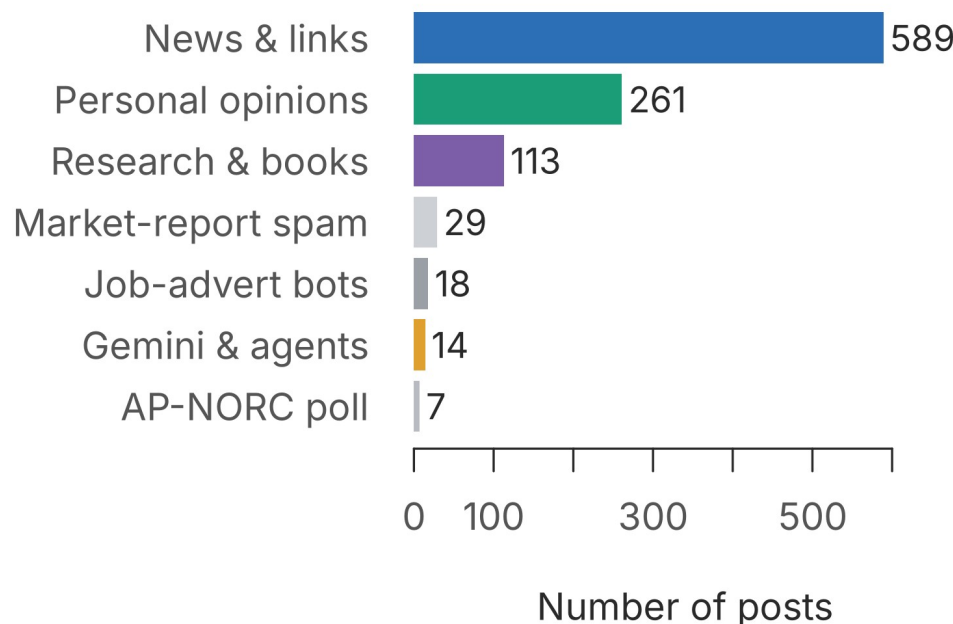
topic clusters

118,988

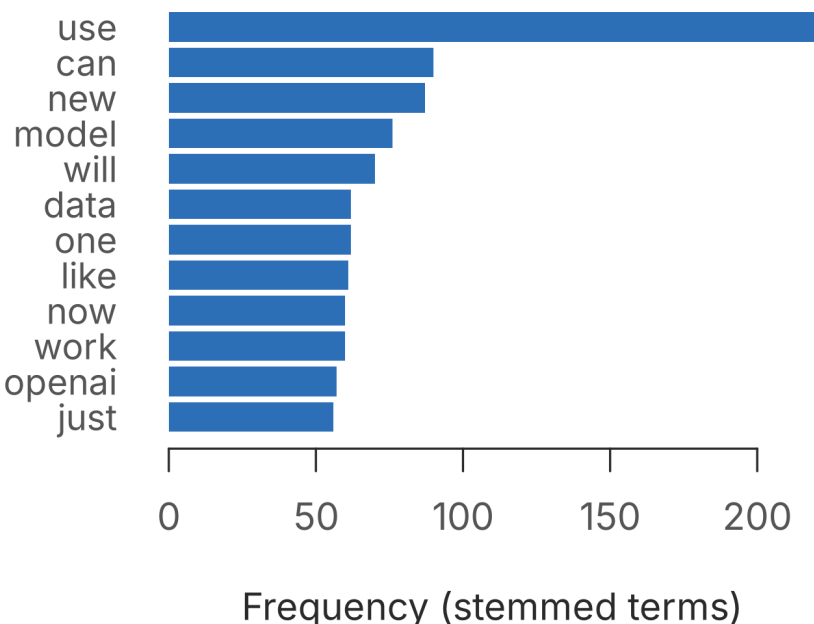
follow edges

1 What do people post about AI?

TF-IDF → cosine distance → MDS → k-means (k = 7)



Posts per topic cluster. Grey = automated or repeated content.



Most frequent stemmed terms: discussion of using AI in practice.

News & link sharing 57%

Personal opinions on AI 25%

These two topics dominate the sample.



3 of 7 clusters are automated or repeated content: job-advert bots, market-report spam and one poll headline posted many times.

2 Which topics get engagement?

Randomisation test for a difference in means (Module 10)

Engagement = likes + reposts + replies

News (n = 589) vs Opinion (n = 261)

H0: equal mean engagement

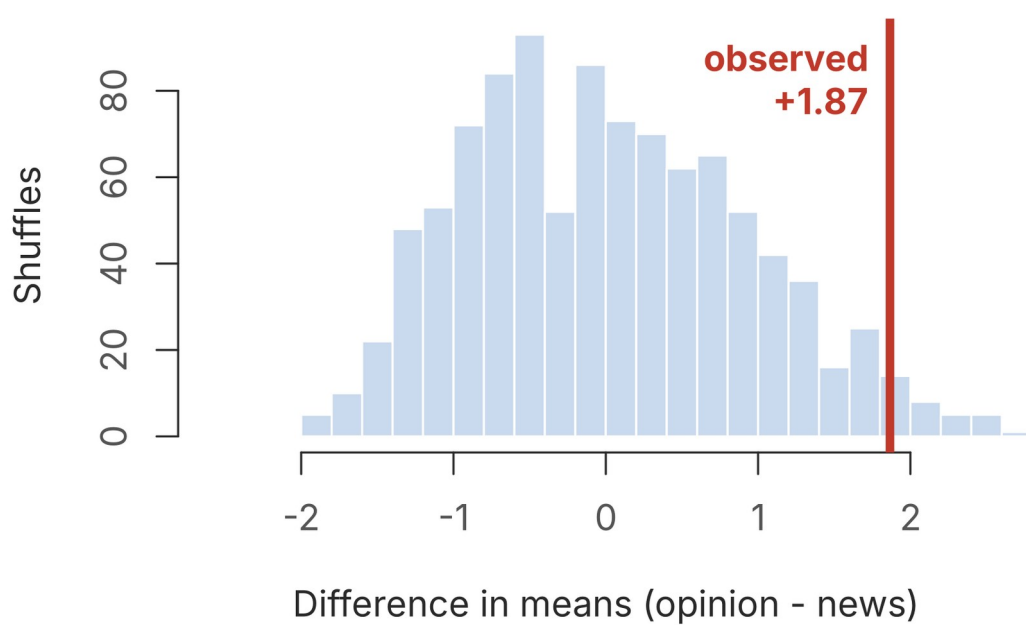
HA: means differ ($\alpha = 0.05$)

p = 0.032

1,000 shuffles · t-test p = 0.046
Reject H0



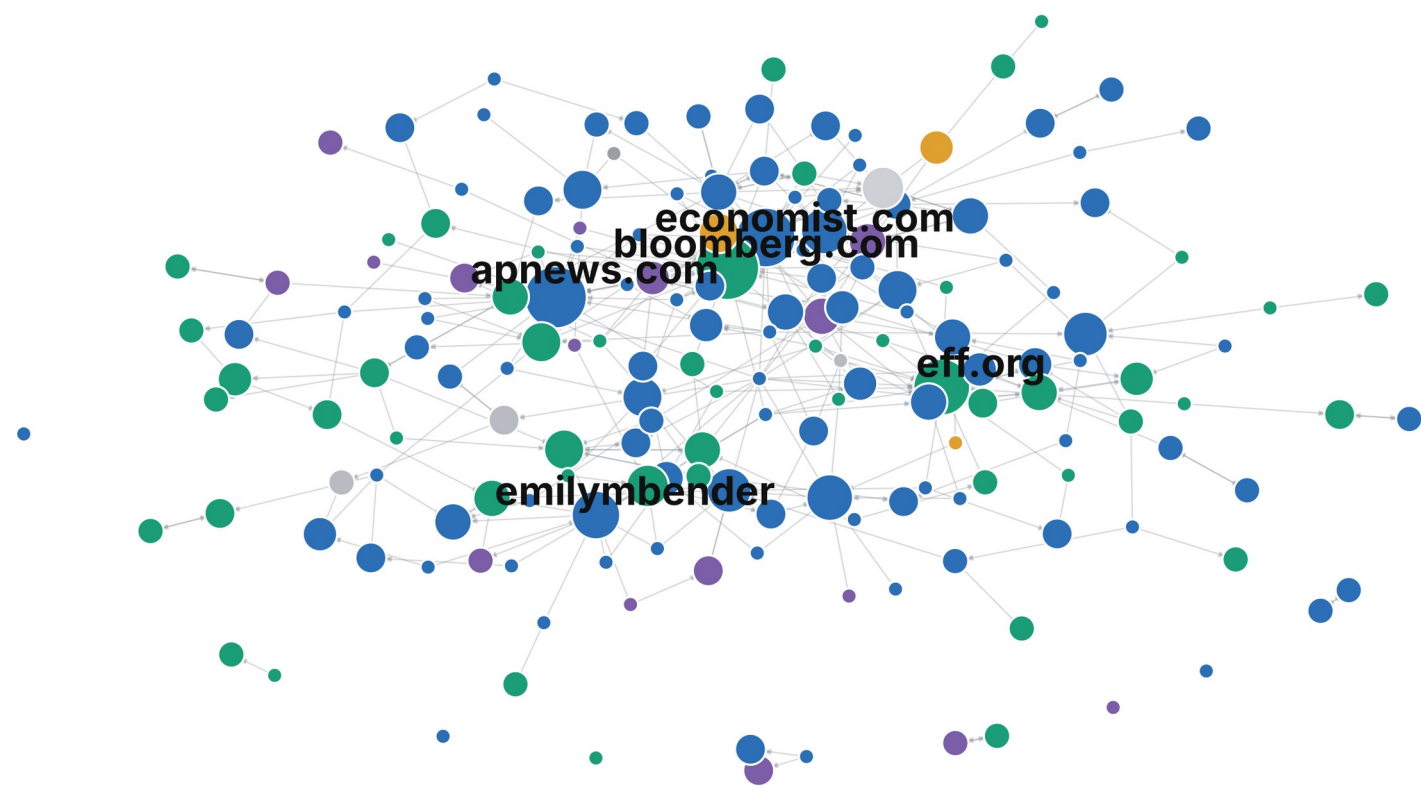
Opinion posts average 4.30 engagements vs 2.43 for news; automated clusters get almost none.



Observed difference (+1.87) lies in the tail of 1,000 label shuffles. Across all 7 clusters, ANOVA p = 0.119.

3 Who is connected, and who is central?

Node = author · edge A → B = A follows B (directed)



News & links Opinions Research Gemini Automated
Authors with degree > 1 (177 nodes, 325 edges); size = in-degree.

340/803

authors with a tie

480

follow edges

0.004

density

18 weak components; the largest holds 294 of the 340 connected authors.

#	In-degree	PageRank	Betweenness
1	apnews.com	ketanjoshi.co	vinkius-mcp-ai
2	bloomberg.com	nposegay	colin-fraser.net
3	eff.org	savasavasava	bigearthdata.ai
4	economist.com	mattseybold	bloomberg.com
5	emilylbender	apnews.com	pauljdavies

Authors mostly follow **established organisations** (news outlets, EFF); **individual commentators** rank highest on PageRank and betweenness.



OVERALL FINDING

AI conversation on Bluesky is mostly news-sharing and personal opinion. Opinion posts receive more engagement than news, and while authors mostly follow established organisations, individual commentators hold the most central positions in the follow network.

Limitations: ~1 day of keyword-search posts, not a random sample · non-English text lost in cleaning · no clear elbow for k · engagement very skewed (median 0) · max ~300 follows per author; a follow is not an interaction.